Negative Lines: Gold Set Reconciliation Reference Manual

Authoritative Reconciliation Protocol, Cause Code Taxonomy & Audit Trail for 10 Gold-Set Patents

Master Reference Guide
OSI Hackathon 2026

Purpose & Objective: To train reliable surrogate models and benchmark extraction agents, a gold-standard dataset must represent factual ground truth rather than algorithmic hallucinations or unverified human guesses. This manual provides an exhaustive reference of how independent human annotations (Labeler B: **Kathe** and **Akhila**) and automated extraction (Labeler A: **Claude AI**) were reconciled across all 10 semiconductor negative photoresist patents, governed by the rules in `hand-labeling-guide.md` and `Human labeling guide and discrepancy list.pdf`.

10	239	1,285	858
PATENTS RECONCILED	FORMULATION RECORDS	CHEMICAL COMPONENTS	LITHOGRAPHY RESULTS

1. THE SIX GOVERNING GOLDEN RULES OF RECONCILIATION

1. Copy, Don't Convert: Raw quantities, units, and bakes are copied verbatim. Units like "1 min" remain 1 min, not 60 s; "0.11 µm" remains 0.11 µm. Normalization to standard units is performed computationally, never by manual guessing.

2. Missing is Null, Never 0: Unstated parameters remain empty (`None`). A zero strictly indicates an empirical measurement finding zero (e.g. 0 nm film loss).

3. Names as Written: Verbatim chemical trade names, codenames (e.g. "PAG 1", "Resin A", "DMC"), and structure labels are preserved. Chemical SMILES and CAS numbers are mapped in separate lookup steps.

4. Provenance Required: Every record and component must possess an exact document location (`location`) and a verbatim text excerpt (`evidence_snippet`).

2. CROSS-PATENT GLOBAL DECISIONS (ITEMS X-1 TO X-5)

X-1. Text vs. Table Contradictions (Team Default): In patents 01, 06, 07, and 08, text and tables disagree. *Resolution:* Table values are adopted as the primary authoritative record; contradictory narrative text is recorded in `labeler_note`; confidence is set to `medium`; cause is tagged as `source_conflict`.

X-2. Coating-Only Formulations: Formulations evaluating only coating uniformity and defectivity without masked lithography (Patents 01, 02, parts of 05, 10). *Resolution:* Retained in the master gold set with `coating_only = True` to support viscosity and spin-coating film thickness surrogate training.

X-3. Group Performance Statements: Global ranges in text (e.g., *"Examples 21-32 resolved lines up to 1.5 µm"*). *Resolution:* Propagated across corresponding records with an explicit `group_statement = True` qualifier.

X-4. Controlled Vocabulary Extensions (vocab.yaml v3): Mapped pinanol/cyclohexanol tosylates to `acid_amplifier`; standardized leveling agents to `surfactant`; classified plasticizers (DOP) as `additive` with plasticizer notes.

X-5. PFAS Compliance Flags: PAGs containing perfluorinated sulfonate anions (triphenylsulfonium triflate, nonaflate, $C_4F_9SO_3^-$) in patents 06, 07, 08, and 09 are explicitly flagged with `PFAS_flag = True` for automated screening by the Compliance Agent.

Taxonomy of Cause Codes & Critical QC Resolutions

Standardized Cause Classification and Case Studies of Resolved Corruptions

Audit & QC Protocol
Cause Codes 1.6

3. THE EIGHT RECONCILIATION CAUSE CODES (SECTION 1.6)

Cause Code	Trigger Condition	Standard Resolution Rule	Representative Example
SOURCE_CONFLICT	Patent text contradicts patent tables or claims.	Adopt Table value; record text value in note; set confidence medium.	US9017923B2: Table 1 PEB 95 °C adopted over text 110 °C.
TYPO	Obvious typographical error in source or file corruption.	Record intended value; document error; correct software serial bugs.	KR20240036456A: Excel date serials 46054 / 46259 fixed to 1.2 g / 25.8 g.
BOUNDARY	Records split differently (e.g. multi-dose / multi-film).	Split into separate records per unique condition to preserve 1:1 fidelity.	US20080241753A1: Split into _50nm and _100nm records per example.
METRIC_DEF	Ambiguity in lithographic metric definition.	Standardize to vocab.yaml v3 metric; preserve original text label.	US7432034B2: LER (3σ linewidth) standardized to lwr_3sigma_nm.
BASIS	Ambiguity in quantity basis (phr, wt% solids, wt% total).	Analyze denominator. If resin ≠ 100, use parts_by_mass, not phr.	US20020068237A1: Whole solution amounts assigned parts_by_mass.
GUIDE_UNCLEAR	Labeling guide or vocabulary did not clearly specify role.	Update mapping per vocab.yaml v3; log rule clarification.	US20210018838A1: Leveling agent mapped to role surfactant.
MISSED	Component, bake parameter, or result omitted in one pass.	Verify against source PDF and reinstate in reconciled record.	CN120370628A: Added developing time 60 s from claim 9.
UNIT	Unit transcription discrepancy or missing printed unit.	Preserve printed unit verbatim; leave blank if omitted (never guess).	WO2004107052A1: Table 2 surfactant content kept unitless.

4. HIGH-IMPACT QUALITY CONTROL (QC) CASE STUDIES

Case Study 1: Resolving the Excel Date-Serial Corruption (KR20240036456A)

The Problem: In the human model handoff sheet (`QC Issues`), 58 rows were flagged with `DATE_SERIAL_CORRUPTION`, where Irgacure 369 appeared as `46054` and PGMEA appeared as `46259`. This occurred because automated Excel re-saving formatted strings like "1.2" and "25.8" into dates (e.g. 1/2/2026).

Reconciliation Resolution: Cross-checked against Kathe's independent annotations and the Korean full text. Conclusively confirmed ground-truth quantities across Examples 1–5: **Novolac = 10.0 g**, **DPHA = 6.0 g**, **Irgacure 369 = 1.2 g**, and **PGMEA = 25.8 g**. Cause code: `typo`.

Case Study 2: The Megajoule Dose Header Typo (US20070248896A1 Table III)

The Problem: Table III prints exposure dose with header unit `MJ/cm2` (Megajoules per square centimeter). A megajoule dose would vaporize a silicon wafer.

Reconciliation Resolution: Paragraph [0035] specifies typical exposure energy between "500 to 2500 mJ". Reconciled unit to **mJ/cm²** with an explicit source typo note in `labeler_note`. Cause code: `typo` / `unit`.

Case Study 3: The Text-vs-Table Thickness Reversal (US20080241753A1)

The Problem: Paragraphs [0158] and [0163] state that Example 1 was tested at 50/100 nm and Example 2 at 100/200 nm. However, Table 2 in the same patent inverts this, assigning 100/200 nm to Example 1 and 50/100 nm to Example 2.

Reconciliation Resolution: Enforced Team Decision X-1: adopted **Table 2** values as primary ground truth, documented narrative text inversion in `labeler_note`, and set confidence to medium. Cause code: `source_conflict`.

Patent Resolution Matrix: Patents 01 to 05

Itemized Discrepancy Decisions, Value Reconciliations and Provenance Notes

Audit Trail (Part 1)
Patents 01 – 05

Patent 01: US20210018838A1 · Jiangsu Aisen Semiconductor

6 RECORDS · 24 COMPONENTS · 48 RESULTS

01-2 Bake Conflict: [0023] describes wafer pre-treatment at 120 °C 10m; [0025] specifies resist softbake at 110 °C 5m. *Reconciled:* Softbake = 110 °C 5m; pre-treatment recorded in substrate field. (SOURCE_CONFLICT)

01-3 Thickness Target vs Measured: Text says 110 µm nominal coating; Table 2 measures 49–66 µm. *Reconciled:* Nominal thickness left empty; measured values recorded in Results sheet. (METRIC_DEF)

01-5 Leveling Agent Loading: Table 1 lists leveling agent in ppm (100–1000 ppm) over 100 wt% base. *Reconciled:* Role standardized to `surfactant` per vocab.yaml v3; basis = `ppm_unspecified`. (GUIDE_UNCLEAR)

01-6 Sensitizer Role: Chemically a sulfonium SbF6 + DNQ mixture. *Reconciled:* Assigned role `sensitizer` per vocab v3; full salt name preserved. (METRIC_DEF)

Patent 02: WO2004107052A1 · ADMS Technology (Spinless Coating)

51 RECORDS · 324 COMPONENTS · 204 RESULTS

02-1 Solvent Codenames: DMC, NBA, MCA, CHX, PGPE undefined in text. *Reconciled:* Verbatim codes preserved with confidence=low per Golden Rule 4. (GUIDE_UNCLEAR)

02-3 Surfactant Content: Table 2 surfactant values (0.01, 2) lack printed units. *Reconciled:* Unit left empty; basis = `unspecified`. (UNIT)

02-4 Embodiments 20–51 Solvent: Text confirms DMC is the sole solvent and Table 2 surfactant replaces base R08. *Reconciled:* Single-solvent formulation aligned across all 32 embodiments. (MISSED)

02-5 Coating-Only Flag: No exposure or development. *Reconciled:* Formulations retained; flagged `coating_only = True` per Decision X-2. (BOUNDARY)

Patent 03: CN120370628A · Xi'an Manareco New Materials

21 RECORDS · 73 COMPONENTS · 71 RESULTS

03-1 Missing Translation Tables: Tables 1 & 2 omitted in English Google Patents copy. *Reconciled:* Example 21 (Recipe W1) validated as high-confidence ground truth (Resin A 30g, Resin B 1g, PAG 5g, Cyclopentanone 55g). Examples 22–32 retained with `table_missing_in_source_copy` and confidence=low. (BOUNDARY)

03-6 Post-Bake: "Baked and photo-cured at 120 °C for 60 s" post-exposure. *Reconciled:* Mapped to `peb_temp = 120`, `peb_time = 60`. (METRIC_DEF)

03-7 Developing Time: 60 s in 2.38% TMAH specified in claim 9. *Reconciled:* Recovered from claim and populated into process conditions. (MISSED)

Patent 04: US20020068237A1 · Genji Imai (Visible Light Resists)

6 RECORDS · 32 COMPONENTS · 14 RESULTS

04-1 Whole Mixture Basis: [0151] amounts are parts by weight of whole mixture; resin is 100–167 parts. *Reconciled:* Basis = `parts_by_mass` (not phr). (BASIS)

04-3 Solvent Separation: Melamine varnish solvent (xylol/butanol 50/50). *Reconciled:* Split into explicit solvent rows to ensure downstream surrogate compatibility. (BOUNDARY)

04-5 Acid Amplifier Role: Photoacid proliferating agent pinanol tosylate. *Reconciled:* Assigned controlled role `acid_amplifier` per vocab.yaml v3. (GUIDE_UNCLEAR)

04-6 Visible Exposure: Filtered metal halide lamp (≥480 nm). *Reconciled:* `exposure_source = visible`, `wavelength_min_nm = 480`. (METRIC_DEF)

Patent 05: KR20240036456A · Novolac Dry Film Resists

8 RECORDS · 31 COMPONENTS · 32 RESULTS

05-1 Cut-off PDF Margins: Scanned PDF clipped right-hand quantities. *Reconciled:* Cross-verified against Google Patents uncut HTML full text. (GUIDE_UNCLEAR)

05-2 Date Serial Corruption Fix: Excel date corruptions 46054 / 46259. *Reconciled:* Irgacure 369 = **1.2 g**, PGMEA = **25.8 g** across all examples. (TYPO)

05-8 Monomer Ratios: Table 1 row 2 ratios (1/0.61/0.4) are novolac synthesis ratios. *Reconciled:* Populated into `polymer_monomer_ratio`, not recipe amounts. (BASIS)

Patent Resolution Matrix: Patents 06 to 10

Itemized Discrepancy Decisions, Value Reconciliations and Provenance Notes

Audit Trail (Part 2)
Patents 06 – 10

Patent 06: US20080241753A1 · Tokyo Ohka (Silsesquioxane e-beam)

49 RECORDS · 294 COMPONENTS · 147 RESULTS

06-1 Text vs Table Inversion: Text and Table 2 invert thickness assignments. *Reconciled:* Table 2 values adopted per Decision X-1. (SOURCE_CONFLICT)

06-3 Dual-Thickness Record Splitting: Each example evaluated at two thicknesses. *Reconciled:* Split into distinct records (`_50nm`, `_100nm`). (BOUNDARY)

06-9 Crosslinker Loading: Examples 13–14 crosslinker printed as 12 parts. *Reconciled:* Preserved 12 as printed (verified intentional). (TYPO)

06-11 PFAS Anion Flag: PAG1 (triflate) and PAG2 (nonaflate). *Reconciled:* Explicitly tagged `PFAS_flag = True` for compliance screening. (GUIDE_UNCLEAR)

Patent 07: US7432034B2 · Fujifilm (e-beam & X-ray)

17 RECORDS · 136 COMPONENTS · 85 RESULTS

07-1 Resin Mw/Mn Discrepancy: Synthesis text says Mw/Mn 2.2; Table 1 says 1.2. *Reconciled:* Table 1 value 1.2 adopted per Decision X-1. (SOURCE_CONFLICT)

07-3 Crosslinker Typo: Example 10 lists crosslinker 'C-3'. *Reconciled:* Preserved as printed with note 'likely typo for CL-3'. (TYPO)

07-5 LER vs LWR Metric: Evaluated as 3σ of linewidth. *Reconciled:* Standardized to `lwr_3sigma_nm` with source wording documented. (METRIC_DEF)

07-11 PFAS Flag: PAG I-1 (nonaflate) and PAG4-36 ($C_4F_9SO_3^-$). *Reconciled:* Tagged `PFAS_flag = True`. (GUIDE_UNCLEAR)

Patent 08: US9017923B2 · Shin-Etsu Chemical (Ferrocene Polymers)

16 RECORDS · 48 COMPONENTS · 32 RESULTS

08-1 PEB Temperature: Text says PEB 110 °C 90s; Table 1 specifies 95 °C. *Reconciled:* Table 1 value 95 °C adopted per Decision X-1. (SOURCE_CONFLICT)

08-3 Polymer-Bound PAG: Polymer 6 contains chemically bound acid generator. *Reconciled:* Verified no separate PAG component exists. (GUIDE_UNCLEAR)

08-7 Polymer Etch Rates: Table 2 lists dry etch rates of polymer-only films. *Reconciled:* Maintained as distinct records with `etch_rate_only` flag. (BOUNDARY)

Patent 09: US20230314945A1 · Tokyo Ohka (EUV Silicon Resists)

19 RECORDS · 133 COMPONENTS · 95 RESULTS

09-1 Quencher Quantity Typo: Table 2-cont. lists quencher '(D)-3 [35] 3'. *Reconciled:* Corrected to 35 parts matching series. (TYPO)

09-6 Single Pitch Resolution: Only 16 nm line / 32 nm pitch tested. *Reconciled:* `resolution_min_cd_nm = 16`, condition 'smallest tested'. (METRIC_DEF)

09-9 Photodecomposable Base: Sulfonium salicylates (D)-x acting as base. *Reconciled:* Mapped to role `quencher`. (METRIC_DEF)

Patent 10: US20070248896A1 · FormFactor (Modified SU-8)

46 RECORDS · 240 COMPONENTS · 192 RESULTS

10-1 Chemical Name Typo: Example 3 prints 'GPIMS' instead of GPTMS. *Reconciled:* Preserved GPIMS as printed with GPTMS note. (TYPO)

10-2 Dose Unit Typo: Table III prints dose as 'MJ/cm2'. *Reconciled:* Corrected to mJ/cm² based on text range [0035]. (TYPO)

10-5 Exact Shear Numbers: Text rounds die shear values; Tables I–II give exact numbers. *Reconciled:* Adopted exact table values. (SOURCE_CONFLICT)

10-11 Cyracure 6974 Role: Termed photoinitiator in text, but is SbF6 sulfonium salt. *Reconciled:* Mapped to role `pag`. (GUIDE_UNCLEAR)

Handoff to Multi-Agent Pipeline: The reconciled dataset eliminates unit discrepancies and corruptions, providing the **Surrogate Model** with clean numerical vectors ($R^2 > 0.85$ expected on dose/thickness regressions) and the **Compliance Agent** with structured SMILES/CAS inputs to verify PFAS-free status before design recommendations are finalized.